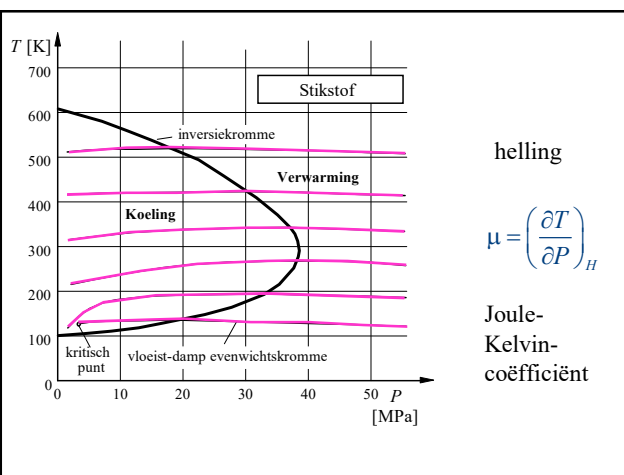
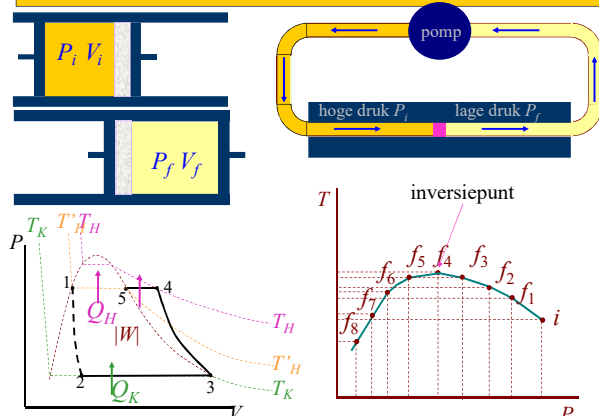


HOOFDSTUK IX

Thermodynamische functies en vergelijkingen

9.1 Enthalpie en het smoorproces



Enthalpie	Inwendige energie
$dH = dQ + VdP$ $\left(\frac{\partial H}{\partial T} \right)_P = C_P$ isobaar proces $\Delta H = Q$ adiabatisch proces $\Delta H = \int_i^f VdP$ smoorproces: $H_i = H_f$ ideaal gas: $dH = C_P dT$	$dU = dQ - PdV$ $\left(\frac{\partial U}{\partial T} \right)_V = C_V$ isochoor proces $\Delta U = Q$ adiabatisch proces $\Delta U = - \int_i^f PdV$ vrije expansie: $U_i = U_f$ ideaal gas: $dU = C_V dT$

$$\begin{aligned}
 dU &= TdS - PdV \\
 dH &= TdS + VdP \\
 dF &= -SdT - PdV \\
 dG &= -SdT + VdP
 \end{aligned}$$

Eerste wet + definities → 4 differentieële basisvergelijkingen voor *reversibele* processen

$$\begin{aligned}
 dU &= TdS - PdV \\
 dH &= TdS + VdP \\
 dF &= -SdT - PdV \\
 dG &= -SdT + VdP
 \end{aligned}$$

$$\begin{aligned}
 U(S,V) \\
 H(S,P) \\
 F(T,V) \\
 G(T,P)
 \end{aligned}$$

$$\begin{aligned}
 \left(\frac{\partial T}{\partial V}\right)_S &= -\left(\frac{\partial P}{\partial S}\right)_V \\
 \left(\frac{\partial T}{\partial P}\right)_S &= \left(\frac{\partial V}{\partial S}\right)_P \\
 \left(\frac{\partial S}{\partial V}\right)_T &= \left(\frac{\partial P}{\partial T}\right)_V \\
 \left(\frac{\partial S}{\partial P}\right)_T &= -\left(\frac{\partial V}{\partial T}\right)_P
 \end{aligned}$$

Maxwellvergelijkingen
bij evenwichtstoestanden

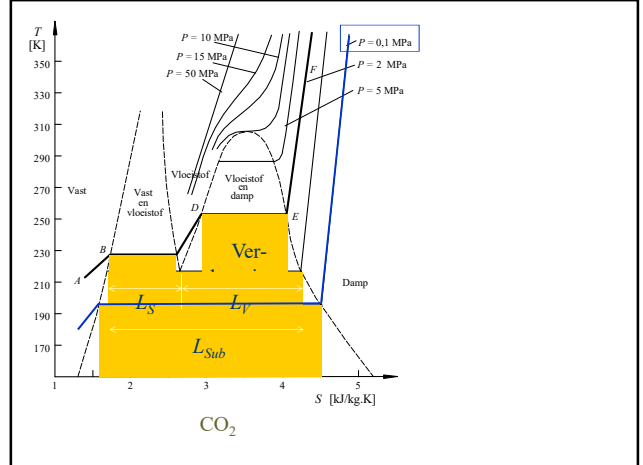
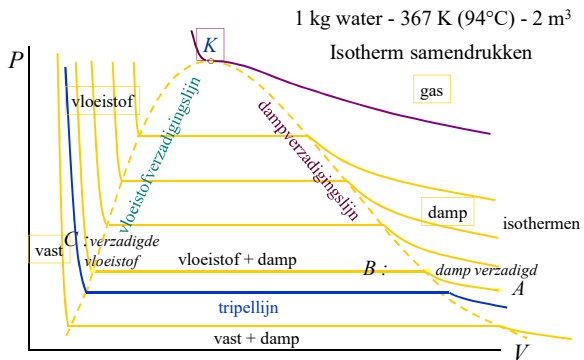
TdS vergelijkingen

$$\begin{aligned}
 TdS &= C_V dT + T\left(\frac{\partial P}{\partial T}\right)_V dV \\
 TdS &= C_P dT - T\left(\frac{\partial V}{\partial T}\right)_P dP
 \end{aligned}$$

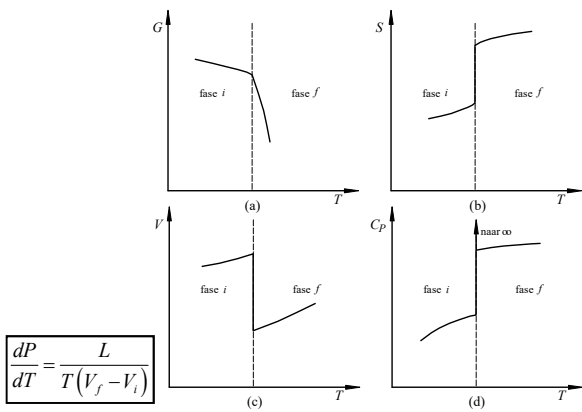
Energievergelijkingen

$$\begin{aligned}
 \left(\frac{\partial U}{\partial V}\right)_T &= T\left(\frac{\partial P}{\partial T}\right)_V - P \\
 \left(\frac{\partial U}{\partial P}\right)_T &= -T\left(\frac{\partial V}{\partial T}\right)_P - P\left(\frac{\partial V}{\partial P}\right)_T
 \end{aligned}$$

3.1 PV-diagram van een zuivere stof



1ste orde fase-transities



2de orde fase-transities

